

WEEK-1

OUT-PUT EXERCISE-3

```
"C:\Program Files\Java\jdk-21\bin\java.exe" "-javaagent:E:\IntelliJ2026\IntelliJ IDEA 2026.1.2\lib\idea_rt.jar=5879
Bubble Sort
103 Kaira 1500.0
101 Rahul 2500.0
104 Priya 5000.0
102 Anita 7000.0
105 Kiran 9000.0

Quick Sort
103 Kaira 1500.0
101 Rahul 2500.0
104 Priya 5000.0
102 Anita 7000.0
105 Kiran 9000.0

Process finished with exit code 0
```

Explanation:

Time Complexity Comparison

Algorithm	Best Case	Average Case	Worst Case
Bubble Sort	$O(n)$	$O(n^2)$	$O(n^2)$
Quick Sort	$O(n \log n)$	$O(n \log n)$	$O(n^2)$

Why Quick Sort is Preferred

1. Quick Sort has an **average time complexity of $O(n \log n)$** , making it much faster for large datasets.
2. Bubble Sort requires many unnecessary comparisons and swaps, resulting in **$O(n^2)$** performance.
3. Quick Sort is widely used in real-world applications because it is efficient and scales well with increasing data size.
4. Bubble Sort is mainly used for educational purposes or very small datasets due to its simplicity.

Conclusion: For an e-commerce platform handling thousands of customer orders, **Quick Sort is generally preferred over Bubble Sort because it provides significantly better performance and faster execution times.**